The Common-Source Trap

Endogenous Independent Evidence in Distributed Discovery

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Abstract

When several searchers can use a free common signal or purchase independent signals, the group may remain concentrated on the common source even though one independent source raises net discovery. This paper studies that wedge in a finite, exact source-choice model. For arbitrary finite team size $N \geq 2$, it derives each common-to-independent payoff threshold, proves that the thresholds are strictly decreasing, and characterizes every weak pure source-count equilibrium. The planner thresholds are also explicit. At the all-common boundary, an equilibrium/planner wedge exists exactly for $p(1-p)(N-1)/N \leq c < p(1-p)$, an interval of width $p(1-p)/N$. Exact enumeration for $N = 2, \ldots, 8$ reproduces the formulas. The result is not a universal under-acquisition theorem: at $N = 3$, $p = 4/5$, and $c = 13/375$, the unique equilibrium purchases two independent sources while the planner purchases one. The paper uses that counterexample to separate source diversity, action diversity, disclosure, and rewards. It relates the theorem to bounded mechanism and architecture results and states experimental predictions without claiming human evidence. Every numerical table and figure is generated from checksum-validated immutable artifacts.

Keywords: information acquisition; strategic experimentation; source diversity; collective search; public goods; distributed discovery.

Evidence status. The central threshold ordering and all-common interval are analytic theorems (DD-C-0057). The $N = 2, \ldots, 8$ tables are exact finite classifications (DD-C-0052). The interior over-acquisition fixture is an exact negative result (DD-C-0058). DD-011 contributes synthetic predictions only; no participant was recruited and no human experiment was conducted.

1 Introduction

Collective search often begins with a source decision. A research team chooses which datasets, assays, experts, or models to consult before it chooses which hypotheses to test. An investment team chooses whether to reuse a familiar diligence channel or commission an independent view before it allocates capital. A multi-agent system chooses whether several agents should retrieve the same context or obtain conditionally independent evidence before they act. These examples share a basic tension. A common source is cheap and makes individual beliefs mutually predictable. An independent source is costly but can create a new route to the target. The benefit of that new route is partly public, whereas its cost is borne by the user who acquires it.

That tension has deep prior art. The distinction between private and social value of information is classical [Hirshleifer, 1971]. In strategic experimentation, information produced by one player's risky action is a public good, creating free-rider and encouragement effects [Bolton and Harris, 1999, Keller et al., 2005]. Coordination motives can make agents want to know what others know, while strategic substitution can instead favor differentiated information [Hellwig and Veldkamp, 2009]. Richer

Table 1: Evidence layers used in this paper. Aggregation does not promote a claim's status.

Object	Result	Evidence category	Boundary
DD-008	Two-agent trap interval	exact bounded grid	one frozen fixture family
DD-008A	$N = 2, \dots, 8$ source counts	exact finite census	registered rational grid
DD-008B	decreasing thresholds and count characterization	analytic theorem	frozen homogeneous model
DD-008B	interior over-acquisition witness	exact negative result	one rational fixture
DD-006B	16 strict mechanism rows	exhaustive exact class	subsidized, bounded transfers
DD-009	20 coherent architectures	bounded exact atlas	288 declared cells only
DD-011	design and power rows	synthetic estimate	no human experiment

Source runs: `20260721T141527Z_DD-008_0d11dc77_7e0c8f1d66`, `20260721T163030Z_DD-008A_8b70668b_06307caab4`, `20260721T192412Z_DD-008B_649deb08_29dbeaf3a9`, `20260721T165512Z_DD-006B_f022a1a5_3be21d0b9b`, `20260721T171249Z_DD-009_bc78d249_0c3851c41a`, `20260721T185647Z_DD-011_fa0271d9_fcaa647c55`; claims: DD-C-0051, DD-C-0052, DD-C-0053, DD-C-0054, DD-C-0056, DD-C-0057, DD-C-0058; generator: `distributed_discovery.papers.build_common_source_trap`; input SHA-256 prefixes: `5e1da8ec2e34`, `6df75b7bf2b1`, `a9d7c5143d75`, `a877471e9075`, `34542e79cb88`, `acf90df6a2dc`, `0d586f4001c8`.

models connect endogenous attention to source clarity, action complementarity, and welfare [Myatt and Wallace, 2012, Colombo et al., 2014]. Costly social learning can strengthen herding [Ali, 2018]; complementary correlated sources can generate long-run learning traps [Liang and Mu, 2020]. The broad phenomenon of acquisition externalities is therefore not new.

The contribution here is narrower and exact. We isolate a one-shot finite model in which N agents select a free shared signal or a costly conditionally independent signal, then search the signaled target directly. Correct searchers split a unit prize. The planner values the union event that at least one searcher is correct, minus acquisition costs. This creates a transparent mapping from source choice to discovery coverage and permits exact type-payoff accounting. The model removes social histories, dynamic experimentation, arbitrary signal design, and strategic action choice. Those omissions make the theorem auditable; they also define its limits.

The first result is a general finite-team threshold characterization. If k agents currently use independent sources, a common user compares its shared prize to the prize it would earn as the $(k+1)$st independent user. The gross gain is

$$A_k = p(1-p)\mathbb{E}\left[\frac{1}{1+X_k} - \frac{1}{N-k+X_k}\right], \qquad X_k \sim \text{Binomial}(k,p).$$

The sequence $A_0, A_1, \dots, A_{N-1}$ is strictly decreasing for every finite $N \geq 2$ and $0 < p < 1$. Consequently a count k is a weak pure equilibrium if and only if $A_k \leq c \leq A_{k-1}$, omitting the nonexistent boundary inequality. The planner's corresponding gross thresholds are $B_k = p(1-p)^{k+1}$ through $k = N-2$, followed by $B_{N-1} = 0$.

At $k = 0$, the comparison becomes especially clear:

$$A_0 = p(1-p)\frac{N-1}{N} < p(1-p) = B_0.$$

Thus all-common source use can remain an equilibrium when every planner optimum uses at least one independent source. The exact interval has width $p(1-p)/N$. This generalizes the two-agent Common-Source Trap while preserving the earlier bounded $N = 2, \dots, 8$ census as a separate computational result.

The second result limits the interpretation. The private and planner threshold sequences are each decreasing, but they need not be globally ordered. For $N = 3$ and the transition from one to two independent sources, $A_1 = p(1-p)(3-2p)/6$ exceeds $B_1 = p(1-p)^2$ exactly when $p > 3/4$. At the

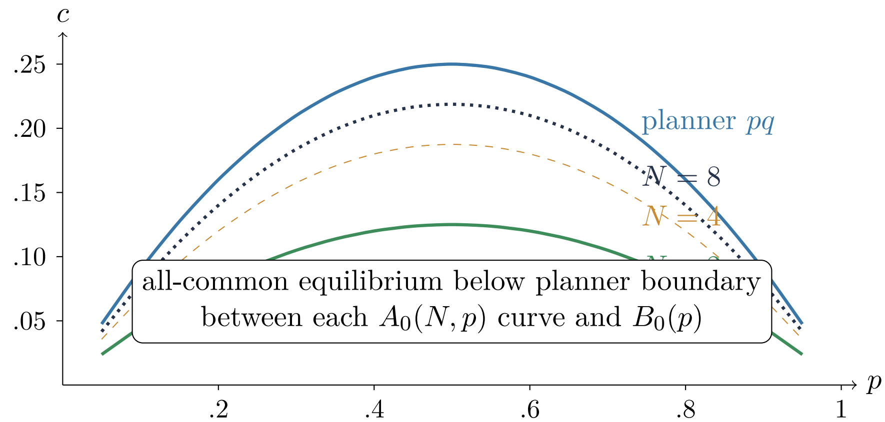

Figure 1: The exact all-common trap region. The upper curve is the planner's first-source threshold $B_0 = p(1-p)$; lower curves are private thresholds $A_0 = p(1-p)(N-1)/N$. The vertical width is $p(1-p)/N$.

Source runs: `20260721T141527Z_DD-008_0d11dc77_7e0c8f1d66,20260721T192412Z_DD-008B_649deb08_29dbeaf3a9`; claims: DD-C-0051, DD-C-0057; generator: `distributed_discovery.papers.build_common_source_trap`; input SHA-256 prefixes: `5e1da8ec2e34, a9d7c5143d75`.

rational fixture $(p, c) = (4/5, 13/375)$, private competition produces two independent sources and the planner wants only one. This exact negative result rules out a paper-wide claim that decentralized teams always under-acquire independence.

The broader lesson is architectural. Source independence changes what evidence exists. Disclosure changes who sees it. Allocation changes whether different signals become different searches. Rewards change private incentives over both reports and actions. A policy that repairs one margin can leave another margin unchanged, and a source-diverse architecture can still collapse into a single action after pooling. The paper therefore connects the source theorem to the project's bounded mechanism, architecture, and experimental-design evidence without treating those distinct models as one welfare theorem.

The remainder proceeds as follows. Section 2 positions the result against the acquisition, learning, experimentation, and cognitive-labor literatures. Section 3 defines the model and accounting. Sections 4–8 present the two-agent result, finite census, general theorem, planner comparison, and exact gap tables. Sections 9–12 separate institutional remedies, joint mechanisms, architecture choices, and experimental predictions. Section 13 records limitations, and Section 14 concludes. Appendices collect proofs, finite-grid interpretation, and reproducibility details.

2 Related literature

2.1 Private and social value of information

The starting point is the established distinction between the private return to information and its social return. Hirshleifer [1971] showed that private information can redistribute rents rather than create commensurate social value. Later information-acquisition work formalizes how strategic externalities and the use of information create wedges between equilibrium and efficient acquisition. Colombo et al. [2014], for example, separate inefficiency in acquiring information from inefficiency in using it and show that efficient use does not guarantee efficient acquisition. The current model shares that separation but uses a different objective: one-shot union discovery rather than a Gaussian-quadratic action economy.

Coordination motives also shape which sources agents choose. Hellwig and Veldkamp [2009] show that strategic complementarity can make information choices complementary: agents may prefer to know what others know. With strategic substitutes, differentiated information can instead become more valuable. Myatt and Wallace [2012] study costly attention across signals that differ in accuracy and clarity; stronger action complementarity can reduce the number of attended signals and make acquired information more public. Those models establish that common-source selection is endogenous to the payoff technology. The exact thresholds below are a specialization to equal prize splitting and direct coverage, not a substitute for those general frameworks.

2.2 Strategic experimentation and information public goods

In multi-agent bandit problems, current experimentation produces information that others can use later. Bolton and Harris [1999] identify both a free-rider effect and an encouragement effect: future experimentation by others can reduce one's incentive to experiment, but it can also increase the value of bringing information forward. Keller et al. [2005] find inefficiently low experimentation rates under stationary Markov behavior and show that asymmetric role patterns can improve efficiency. These are dynamic results with continuation values, belief states, and endogenous timing. Our agents receive a single signal and act once. The analogy is the public-good margin of independent evidence, not the equilibrium object or welfare result.

The dynamic literature is nevertheless important for interpretation. A static source subsidy can close a one-shot threshold gap while failing to solve a dynamic experimentation problem. Conversely, role rotation can distribute the burden of experimentation in a repeated setting even when a one-shot model would assign a single independent source. The institutional section treats assignment and rotation as distinct extensions precisely because the present proof contains no repeated-game incentive.

2.3 Costly herding, source choice, and learning traps

Costly information creates a direct motive to free-ride on public histories. Ali [2018] characterizes when costly acquisition still permits complete social learning; whether information can overturn the public history is central. Liang and Mu [2020] allow learners to choose among flexibly correlated sources and characterize efficient aggregation versus long-run learning traps generated by informational complementarities. Their agents arrive sequentially and pass information across periods. The Common-Source Trap studied here has no public history and no asymptotic learning criterion. It is a source-count equilibrium inside a single coverage problem.

This distinction prevents a tempting but invalid inference. Common source use in our model does not imply an incorrect herd. The shared signal is correct with probability p, and the group may rationally accept concentration when independence is expensive. Likewise, the phrase "trap" describes a precise equilibrium/planner wedge, not an absorbing state in a social-learning process. The value of the finite model is that the wedge, its width, and its disappearance at a boundary are all explicit.

2.4 Scientific and functional diversity

The division of cognitive labor is a longstanding topic in philosophy and science studies. Kitcher [1990] analyzes how incentives can distribute scientists across approaches. Weisberg and Muldoon [2009] use epistemic landscapes to model exploration by heterogeneous scientific agents, and Zollman [2010] studies the value of transient diversity. Importantly, Alexander et al. [2015] challenge broad conclusions drawn from particular landscape models and show that optimal search and the benefits of diversity are sensitive to modeling choices. That disagreement supports the evidence policy used here: define the target law, source structure, action map, and objective before claiming that more diversity is better.

Functional diversity can also improve collective problem solving under specific assumptions [Hong and Page, 2004], while coordinated experiment choice can reduce redundant scientific search [Rzhetsky et al., 2015]. Those models involve heuristics, topics, or experiment portfolios. Our source independence is much narrower. Two agents can possess independent evidence yet take the same action; agents can possess the same evidence yet be assigned different actions. Source diversity and action diversity are therefore separate state variables.

Boundary. The literature review supports adjacency statements, not a novelty certificate. Information-acquisition externalities, public-good experimentation, costly herding, source-choice coordination, and cognitive division of labor all predate this paper. The new claim recorded here is only the exact theorem for the declared coverage-and-prize model.

3 Model

There are $N \geq 2$ agents and three possible targets. The target $\theta \in \{1, 2, 3\}$ is uniform. Before observing evidence, each agent chooses one of two source types. The common source C is free. Every agent selecting C observes the same draw. An independent source I costs $c \geq 0$ per user; each I user observes a distinct draw conditionally independent of every other source given θ. Every source has homogeneous accuracy $p \in (0, 1)$:

$$\mathbb{P}(S = \theta \mid \theta) = p, \qquad \mathbb{P}(S = s \neq \theta \mid \theta) = \frac{1-p}{2}.$$

Agents follow their clues directly, so action $a_i = S_i$. There is no report, message, or contingent action policy. If at least one action equals the target, a unit prize is split equally among correct agents. An incorrect agent receives zero. The planner's gross discovery is the probability that the target appears in the action union. Social net value subtracts the real acquisition cost kc when k agents use independent sources.

The source-count representation is lossless because agents of a given type are symmetric. Let $k \in \{0, \ldots, N\}$ denote the number of I users and $m = N - k$ the number of C users. For $k < N$, the group sees $k + 1$ conditionally independent source draws: k individual draws plus one common draw. For $k = N$, it sees N draws. Hence gross discovery is

$$G_N(k, p) = \begin{cases} 1 - (1-p)^{k+1}, & k < N, \\ 1 - (1-p)^N, & k = N. \end{cases} \tag{1}$$

The last independent purchase, from $k = N - 1$ to $k = N$, creates no new source draw; the sole remaining common user was already the sole user of that draw.

Let $X_r \sim \text{Binomial}(r, p)$ count correct independent users among r other agents. A common user's gross payoff at a profile with $k < N$ is

$$C_k = p\mathbb{E}\left[\frac{1}{N - k + X_k}\right]. \tag{2}$$

The common user receives a prize only if the common draw is correct; then all $m = N - k$ common users are correct together and share with X_k correct independent users.

For $0 < k < N$, an independent user's gross payoff is

$$I_k = p\left[(1-p)\mathbb{E}\frac{1}{1+X_{k-1}} + p\mathbb{E}\frac{1}{N-k+1+X_{k-1}}\right]. \tag{3}$$

When the common draw is wrong, the independent winner shares only with correct independent peers. When it is correct, the $N-k$ common users join the prize. At $k=N$, the second case collapses and $I_N = p\mathbb{E}[1/(1+X_{N-1})]$. Net independent payoff is $I_k - c$.

Two accounting checks are useful. First,

$$kI_k + (N-k)C_k = G_N(k,p), \tag{4}$$

with the absent type omitted. The prize paid across correct agents sums to one exactly on the discovery event. Second, direct enumeration over the target and all source-signal states reproduces equations (1)–(4). The immutable DD-008A audit performs that independent check on the registered finite grid (DD-C-0052).

3.1 Equilibrium and planner conventions

A source-count profile is a weak pure equilibrium if no C user gains by becoming an I user and no I user gains by becoming a C user. It is strict if both applicable inequalities are strict. Because individual identities do not affect payoffs, a count classification covers every labeled profile at that count. Ties are retained. At a threshold cost, adjacent source counts can both be weak equilibria.

The planner chooses k to maximize

$$W_N(k,p,c) = G_N(k,p) - kc. \tag{5}$$

Planner ties are also retained. The comparison does not add agents' prize payments to social value; the unit prize is an internal transfer used to induce private behavior. Discovery is the gross social objective and source costs are real resource costs. Alternative accounting would define a different model.

4 The two-agent Common-Source Trap

With two agents there are three source counts. At $k=0$, common/common discovery is p. A unilateral independent purchase creates a second conditionally independent draw, so discovery becomes $2p-p^2$ and social net value becomes $2p-p^2-c$. The planner therefore prefers one independent source exactly when

$$c < p(1-p). \tag{6}$$

Private incentives are smaller at the all-common profile. Each common user receives $p/2$. After deviating, the buyer is correct with probability p; it receives the whole prize when the common draw is wrong and half when the common draw is correct. Its gross payoff is

$$p(1-p) + \frac{p^2}{2}.$$

The deviation gain before cost is $p(1-p)/2$. Thus common/common remains a weak equilibrium exactly when

$$c \geq \frac{p(1-p)}{2}. \tag{7}$$

$N\backslash c$	0	1/24	1/12	1/8	1/6	1/4
2	1	1	1	0	0	0
3	2	2	1	1	0	0
4	3	2	2	1	0	0
5	4	3	2	1	1	0
6	5	3	2	1	1	0
7	6	4	2	1	1	0
8	7	4	2	1	1	0

Figure 2: Minimum weak-equilibrium independent-source count on the registered $p = 2/3$ DD-008A grid.

Source runs: `20260721T163030Z_DD-008A_8b70668b_06307caab4,20260721T192412Z_DD-008B_649deb08_29dbeaf3a9`; claims: DD-C-0052, DD-C-0057; generator: `distributed_discovery.papers.build_common_source_trap`; input SHA-256 prefixes: `6df75b7bf2b1, a9d7c5143d75`.

Combining (6) and (7) yields the exact two-agent trap interval

$$\frac{p(1-p)}{2} \leq c < p(1-p). \tag{8}$$

DD-008 evaluates a registered rational grid and checks the formulas by direct target/signal enumeration. Five of its fifteen parameter cells fall in the interval (DD-C-0051). That count is a property of the selected grid, while equation (8) is the analytic boundary for the frozen two-agent model. Neither statement estimates how real teams select evidence.

At the canonical illustrative accuracy $p = 2/3$, the private threshold is 1/9 and the planner threshold is 2/9. The registered cost $c = 1/8$ lies between them. Common/common is then a strict source-choice equilibrium with discovery 2/3, whereas one independent source yields discovery 8/9 and net value $55/72 > 2/3$. The wedge is not caused by mistaken Bayesian updating: both source laws and payoffs are common knowledge. It comes from prize sharing. The buyer pays the full source cost but captures only part of the new union probability.

The name "Common-Source Trap" refers to this exact coexistence of a common- source equilibrium and a planner preference for at least one independent source. It does not mean that a common source is intrinsically bad. When cost is high, all-common can be both equilibrium and planner optimal. When cost is low, private users acquire independence. If robustness, common calibration, or coordination has value outside the target-union objective, the planner boundary would also change.

5 Finite-team classification

DD-008A extends the exact accounting to $N = 2, \ldots, 8$, accuracies $p \in \{1/2, 2/3, 3/4\}$, and six rational costs. It evaluates 126 (N, p, c) rows and every source count in each row. Closed-form binomial payoffs are compared with a distinct direct enumerator over targets and source signals. The two implementations agree in every cell, including prize conservation, equilibrium status, and planner values (DD-C-0052).

The census preceded the general proof and remains useful for three reasons. First, it is a regression suite for the payoff formulas. Second, it exposes ties that a generic parameter argument can obscure. Third, it records how the equilibrium and planner threshold crossings interact on a policy-relevant but explicitly bounded rational grid. It is not the basis for the theorem's unrestricted finite-N quantifier.

Figure 2 reports the minimum weak-equilibrium count at $p = 2/3$. At zero cost, the minimum is $N - 1$: once all but one users have independent sources, the remaining common source is already

$N\backslash c$	0	1/24	1/12	1/8	1/6	1/4
2	2	1	1	1	1	0
3	3	2	1	1	1	0
4	4	2	1	1	1	0
5	5	2	1	1	1	0
6	6	2	1	1	1	0
7	7	2	1	1	1	0
8	8	2	1	1	1	0

Figure 3: Maximum planner-optimal independent-source count on the same registered grid.

Source runs: `20260721T163030Z_DD-008A_8b70668b_06307caab4`, `20260721T192412Z_DD-008B_649deb08_29dbeaf3a9`; claims: DD-C-0052, DD-C-0057; generator: `distributed_discovery.papers.build_common_source_trap`; input SHA-256 prefixes: `6df75b7bf2b1, a9d7c5143d75`.

an independent draw and switching the last user is payoff-neutral. Positive costs reduce equilibrium acquisition. At larger N, however, the count need not fall monotonically for a fixed cost in a way that can be summarized by the all-common boundary alone; interior prize competition matters.

The planner counts in Figure 3 follow the declining marginal discovery increments. At zero cost, $N-1$ and N tie, and the displayed maximum is N. At any positive cost, $k = N$ is strictly dominated by $N-1$ because both have gross discovery $1-(1-p)^N$ but the former pays one additional cost. This terminal redundancy is a consequence of defining a "common source" by its shared draw rather than by a fixed provider label.

The tables also show why a single statistic such as "number of independent sources" is insufficient for welfare interpretation. A larger equilibrium count may improve discovery but waste cost at the margin. A smaller count may reflect an underprovided public good or an efficient response to expensive signals. Exact threshold comparison is needed before assigning a sign.

6 General finite-team thresholds

We now state the general result established in DD-008B (DD-C-0057). Write $q = 1-p$. Consider a profile with $k < N$ independent users. A common user that switches becomes an independent user in the $k+1$ profile. Let

$$A_k = I_{k+1} - C_k \tag{9}$$

denote the gross payoff gain before paying c. Substituting equations (2) and (3) and coupling the other k independent outcomes gives

$$A_k = pq\mathbb{E}\left[\frac{1}{1+X_k} - \frac{1}{N-k+X_k}\right], \qquad X_k \sim \text{Binomial}(k,p). \tag{10}$$

Theorem 1 (private thresholds and equilibrium counts). For every finite $N \geq 2$ and $0 < p < 1$,

$$A_0 > A_1 > \cdots > A_{N-1} = 0.$$

A count $k \in \{0, \ldots, N\}$ is a weak pure source-count equilibrium if and only if

$$A_k \leq c \leq A_{k-1}, \tag{11}$$

where the left inequality is omitted at $k = N$ and the right inequality is omitted at $k = 0$.

The characterization follows directly from unilateral deviations. At count k, a common user does not switch if $c \geq A_k$. An independent user does not switch back if $c \leq A_{k-1}$, because that reverse deviation compares the same two adjacent source-count profiles. Strict threshold ordering then implies a unique equilibrium count except at a threshold, where two adjacent counts are weak equilibria.

The nontrivial step is monotonicity. Define

$$f_m(x) = \frac{1}{x+1} - \frac{1}{x+m}, \qquad m = N - k.$$

Couple $X_{k+1} = X_k + Y$ with $Y \sim \text{Bernoulli}(p)$ independent. Conditional on $X_k = x$,

$$f_m(x) - \mathbb{E}[f_{m-1}(x+Y)] = \frac{(x+1)(x+2) + p(m-2)(2x+m+1)}{(x+1)(x+2)(x+m-1)(x+m)}. \tag{12}$$

For $m \geq 2$, numerator and denominator are positive. Multiplying by pq and averaging proves $A_k > A_{k+1}$ for $k = 0, \ldots, N-2$. Equation (10) gives $A_{N-1} = 0$.

The proof is analytic. The DD-008B executable audit instantiates 16,368 positive-kernel identities through $N = 32$, matches 84 independently enumerated private and planner margins through $N = 5$, reproduces all 126 immutable DD-008A classifications, and rejects altered threshold and census artifacts. Those checks audit transcription and implementation; they do not replace the positive-kernel proof.

6.1 Boundary and large-team behavior

At $k = 0$, $X_0 = 0$, so equation (10) becomes

$$A_0 = pq\left(1 - \frac{1}{N}\right). \tag{13}$$

For fixed k and increasing N, the second reciprocal term in (10) vanishes. Using

$$\mathbb{E}\frac{1}{1+X_k} = \frac{1-q^{k+1}}{(k+1)p}$$

gives

$$\lim_{N\to\infty} A_k = \frac{q(1-q^{k+1})}{k+1}. \tag{14}$$

At fixed $k = 0$, the private threshold approaches the planner's first-source threshold from below. For interior fixed k, the limit need not lie below the planner margin, foreshadowing the counterexample in Section 7.

7 Planner comparison

From equation (1), the planner's gross gain from the transition $k \to k+1$ is

$$B_k = G_N(k+1,p) - G_N(k,p) = \begin{cases} pq^{k+1}, & k = 0, \ldots, N-2, \\ 0, & k = N-1. \end{cases} \tag{15}$$

The sequence is strictly decreasing through the positive margins. A planner count k is optimal if and only if $B_k \leq c \leq B_{k-1}$, with the same boundary conventions. Thus equilibrium and planner counts are generated by two ordered threshold sequences. The source-choice problem is not computationally hard once those sequences are known; the economic content lies in their differences.

Corollary 1 (general all-common trap). For every finite $N \geq 2$ and $0 < p < 1$, all-common source use is a weak equilibrium while every planner optimum uses at least one independent source exactly when

$$pq\frac{N-1}{N} \leq c < pq. \tag{16}$$

The interval has width pq/N.

The lower endpoint is included because at $c = A_0$ the all-common profile is a weak equilibrium. The upper endpoint is excluded because at $c = B_0$ the planner is indifferent between zero and one independent source; not every planner optimum then uses independence. If one instead asked whether at least one planner optimum uses independence, the endpoint convention would differ. The paper uses the stronger "every planner optimum" statement.

The width shrinks with N, but this does not mean that all private/planner differences vanish in large teams. It only describes the first transition away from the all-common boundary. As N increases at fixed p, A_0 rises toward B_0. A fixed cost strictly below pq therefore eventually falls below the private threshold, making all-common unstable. Interior thresholds can behave differently because prize competition among already independent users changes the private value of an additional source.

7.1 An exact counterexample to universal under-acquisition

For $N = 3$ and $k = 1$, equation (10) simplifies to

$$A_1 = \frac{pq(3-2p)}{6}, \qquad B_1 = pq^2. \tag{17}$$

Their difference changes sign at $p = 3/4$: $A_1 > B_1$ exactly when $p > 3/4$. Set $p = 4/5$ and $c = 13/375$. Then

$$A_1 = \frac{14}{375} > \frac{13}{375} > \frac{12}{375} = B_1.$$

Since the threshold sequences are decreasing, the unique equilibrium count is $k = 2$ and the unique planner count is $k = 1$ (DD-C-0058). Direct target/signal enumeration reproduces both payoffs and rejects a corrupted threshold.

The mechanism is rent competition rather than a discovery externality with a fixed sign. Moving from one to two independent users changes how a prospective buyer shares the prize with common and independent winners. When accuracy is high, avoiding the block of common winners can raise private prize share even after the planner's incremental union probability has become small. The exact example does not show that over-acquisition is typical. It shows that the phrase "the Common-Source Trap" cannot stand for a universal equilibrium ordering at every interior count.

8 Acquisition, discovery, and welfare gaps

The DD-008A census records three related but distinct differences. The *independence gap* compares source counts. The *discovery gap* compares union probabilities. The *welfare gap* compares net planner value with net value at a selected equilibrium count. Conflating these gaps can reverse an interpretation. More independent sources may increase discovery while reducing net value; equal source counts can still support different action technologies in a richer architecture.

Figure 4 uses the DD-008A convention: maximum planner count minus minimum weak-equilibrium count. Positive values highlight a possible under-acquisition comparison. Negative entries are not errors; they preserve interior over-acquisition on the registered grid. At ties, the max/min convention deliberately displays the widest count wedge and should not be confused with an equilibrium-selection prediction.

$N\backslash c$	0	1/24	1/12	1/8	1/6	1/4
2	1	0	0	1	1	0
3	1	0	0	0	1	0
4	1	0	-1	0	1	0
5	1	-1	-1	0	0	0
6	1	-1	-1	0	0	0
7	1	-2	-1	0	0	0
8	1	-2	-1	0	0	0

Figure 4: Registered independence gap: maximum planner count minus minimum weak-equilibrium count.

Source runs: `20260721T163030Z_DD-008A_8b70668b_06307caab4,20260721T192412Z_DD-008B_649deb08_29dbeaf3a9`; claims: DD-C-0052, DD-C-0057; generator: `distributed_discovery.papers.build_common_source_trap`; input SHA-256 prefixes: `6df75b7bf2b1, a9d7c5143d75`.

$N\backslash c$	0	1/24	1/12	1/8	1/6	1/4
2	0	0	0	2/9	2/9	0
3	0	0	0	0	2/9	0
4	0	0	$-2/27$	0	2/9	0
5	0	$-2/81$	$-2/27$	0	0	0
6	0	$-2/81$	$-2/27$	0	0	0
7	0	$-8/243$	$-2/27$	0	0	0
8	0	$-8/243$	$-2/27$	0	0	0

Figure 5: Gross discovery at the maximum planner count minus gross discovery at the minimum weak-equilibrium count. Negative cells are retained.

Source runs: `20260721T163030Z_DD-008A_8b70668b_06307caab4,20260721T192412Z_DD-008B_649deb08_29dbeaf3a9`; claims: DD-C-0052, DD-C-0057; generator: `distributed_discovery.papers.build_common_source_trap`; input SHA-256 prefixes: `6df75b7bf2b1, a9d7c5143d75`.

Discovery is a monotone function of k through $N-1$, but the last transition is redundant. Figure 5 therefore translates count gaps into the target-union objective. A negative entry means the displayed minimum equilibrium count yields greater gross discovery than the displayed maximum planner count; this can occur only because the planner is accounting for cost. It is not a claim that the planner prefers less discovery as an intrinsic good.

The welfare gaps in Figure 6 are weakly nonnegative by construction because the planner count maximizes equation (5). They can be zero even when source counts differ, either because of a tie or because cost and discovery changes exactly offset. This table is the proper place to discuss inefficiency on the registered grid. The count and discovery tables diagnose mechanisms; the welfare table records the objective actually optimized by the planner.

Three reporting choices matter. First, exact fractions are retained rather than rounded decimals. Second, all registered costs remain visible, including null, tied, and negative-gap cells. Third, the grid is not interpolated into an empirical response surface. The general theorem supplies analytic threshold boundaries, while the finite tables serve as auditable examples and regressions.

9 Institutional remedies

An acquisition wedge suggests several possible interventions, but they act on different primitives. A source subsidy lowers the private cost of independence. An assignment rule replaces source choice with an organizational decision. A rotation rule distributes acquisition burden over time. Attribution or priority protection changes the private return to producing evidence. Disclosure changes who

$N\backslash c$	0	1/24	1/12	1/8	1/6	1/4
2	0	0	0	7/72	1/18	0
3	0	0	0	0	1/18	0
4	0	0	1/108	0	1/18	0
5	0	11/648	1/108	0	0	0
6	0	11/648	1/108	0	0	0
7	0	49/972	1/108	0	0	0
8	0	49/972	1/108	0	0	0

Figure 6: Planner net value minus net value at the minimum weak-equilibrium count. The planner definition makes every entry weakly nonnegative.

Source runs: `20260721T163030Z_DD-008A_8b70668b_06307caab4,20260721T192412Z_DD-008B_649deb08_29dbeaf3a9`; claims: DD-C-0052, DD-C-0057; generator: `distributed_discovery.papers.build_common_source_trap`; input SHA-256 prefixes: `6df75b7bf2b1, a9d7c5143d75`.

observes acquired evidence. Allocation changes how evidence becomes search actions. A joint reward can make truth-telling and differentiated action privately attractive. These instruments should not be treated as aliases.

At the all-common boundary, a per-independent-source subsidy s changes the private condition to $c - s < A_0$. The smallest strict subsidy is any $s > c - A_0$ when $c \geq A_0$. If the planner wants the first independent source, $c < B_0$, then the set of subsidies that strictly induce a unilateral purchase is nonempty. This is a local implementation observation, not a budget-balance result. Subsidy finance and selection among multiple buyers are outside the model.

Assignment is mechanically stronger. A planner can designate exactly $k^s tar$ independent users and thereby attain the source-count optimum without changing private source rewards. Yet assignment creates participation and observability questions. Can the organization verify that the source is independent? Can it observe source cost? Is an assigned user compensated? Does the source remain independent after preprocessing or shared prompts? Those questions are absent from the theorem and must be specified in an implementation model.

Priority and attribution can change the private return to evidence production. The science literature emphasizes that incentives help allocate cognitive labor across approaches [Kitcher, 1990]. Experimental work also shows that competition for novelty can change sampling behavior [Tiokhin and Derex, 2019]. But the sign is not automatic. Strong priority rewards can induce duplicated races, premature disclosure avoidance, or the exact kind of interior over-acquisition identified above. An intervention should therefore target a declared estimand rather than "more diversity" in the abstract.

Figure 7 emphasizes sequencing. Acquisition repair creates distinct evidence. Allocation repair turns evidence into distinct actions. A joint mechanism may additionally support truthful reports and obedience. The numbers come from different registered models and are not additive. In particular, 8/9 in the acquisition fixture and 11/12 in the differentiated allocation fixture do not describe successive outcomes of one unified game.

10 Joint truth and allocation mechanisms

Source acquisition is only the first layer of distributed discovery. Once evidence exists, agents may report strategically or ignore recommended actions. DD-006B studies a separate two-agent, three-target mechanism class with conditionally independent 2/3-accurate signals, a normalized grid of score, obedience, and coverage components, and four observability regimes. It exhausts 60 mechanism rows and 57,600 unilateral report/action deviations.

Sixteen rows strictly implement truthful reporting and a differentiated recommendation for both fixed tie roles. The maximum exact all-tie margin is 13/72, and truthful differentiated discovery is

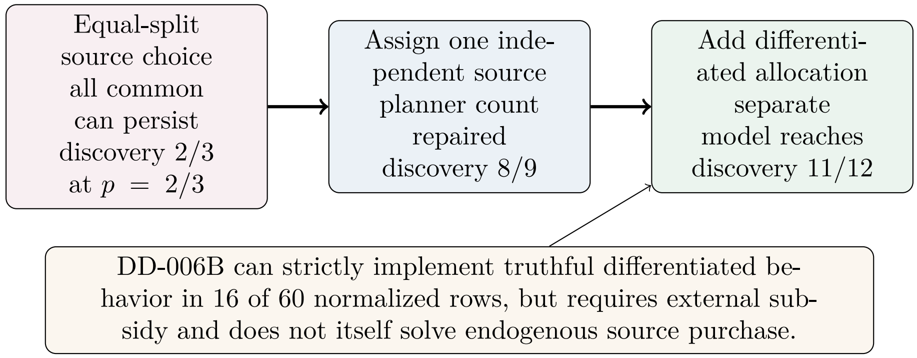

Figure 7: Interventions repair different margins. DD-006B's maximum exact strict margin is 13/72 and best strict discovery is 11/12; those results use a different two-agent observability/transfer model.

Source runs: `20260721T141527Z_DD-008_0d11dc77_7e0c8f1d66,20260721T165512Z_DD-006B_f022a1a5_3be21d0b9b,` `20260721T192412Z_DD-008B_649deb08_29dbeaf3a9`; claims: DD-C-0051, DD-C-0053, DD-C-0057; generator: `distributed_discovery.papers.build_common_source_trap`; input SHA-256 prefixes: `5e1da8ec2e34, 34542e79cb88, a877471e9075`.

11/12 (DD-C-0053). No weak row exists when the target is hidden and actions are observed only through the restricted target-hidden/action-hidden regime. These are exact results for the registered normalized class, not a dominant-strategy theorem or an arbitrary- transfer characterization.

The mechanism is also externally subsidized. Minimum strict expected subsidy in the best regime is positive, and the Architecture Atlas accounts for transfer budgets separately from discovery. Consequently DD-006B cannot be cited as a free cure for the source-choice wedge. It begins after independent evidence is already present and studies incentives to report and act. A complete mechanism would need to combine acquisition, reporting, action, source verification, and budget balance in one game.

This separation creates a useful research agenda. A source subsidy can induce independence but may leave reports untruthful. A truthful scoring component can elicit reports but leave both agents taking the same action. A marginal-coverage component can reward differentiated action but may require target or action observability unavailable in practice. Each result should travel with its information requirements.

The DD-006B result is therefore best read as a feasibility witness. It proves that strict joint truth/allocation incentives are possible in a bounded subsidized class under sufficiently rich observability. It does not prove that a specific organization can implement them, that the mechanism is budget-balanced, or that people will comply. The experimental design in Section 12 treats compliance as an open prediction.

11 Architecture Atlas implications

DD-009 places acquisition inside a bounded Cartesian architecture space. Its declared dimensions are evidence, disclosure, allocation, timing, and reward. Validity rules classify all 288 combinations; 20 coherent cells are evaluated exactly and 268 are rejected with machine-readable reasons. Twelve evaluated cells are nondominated under six fixed objectives. This is a bounded Atlas, not a universal ranking (DD-C-0054).

The acquisition slice in Figure 8 holds two independent channels and information cost 1/4 fixed. Direct private action yields discovery 8/9. Sequential action or planner-style differentiated allocation yields 11/12. Pooling followed by consensus collapses discovery back to 2/3 despite the independent

ID	Disclosure/allocation	Reward	Discovery	Net value	Boundary
A145	private/direct	equal-split	8/9	23/36	protocol/selection
A199	private/sequential	equal-split	11/12	2/3	protocol/selection
A209	private/joint	DD-006A	11/12	2/3	weak mechanism
A210	private/joint	DD-006B	11/12	−13/12	strict subsidized
A229	pooled/consensus	equal-split	2/3	5/12	protocol/selection
A254	pooled/planner	pooling	11/12	2/3	protocol/selection

Figure 8: Acquisition-related slice of the bounded DD-009 Architecture Atlas. Every row uses two independent channels and cost 1/4; changing disclosure, allocation, timing, and rewards changes discovery, net value, and incentive status. This is not a universal architecture ranking.

Source runs: `20260721T171249Z_DD-009_bc78d249_0c3851c41a`; claims: DD-C-0054; generator: `distributed_discovery.papers.build_common_source_trap`; input SHA-256 prefixes: `acf90df6a2dc`.

sources. This is the source/action distinction in its clearest form: acquiring independent evidence is not sufficient if the action protocol compresses it.

Reward changes can also alter accounting without changing discovery. A sole- rescue or marginal-coverage transfer can use budget while preserving the same direct actions. The DD-006A row weakly supports truth and obedience with zero recorded transfer budget under its restricted accounting; the DD-006B row is strict but carries a large external budget. Those rows are not ordered by one score. The Atlas retains discovery, social net value, transfer budget, truthfulness, obedience, action quality, and operating quantities separately.

The Atlas also prevents favorable cherry-picking. A claim that "independent evidence plus pooling works" must specify the allocation rule. A claim that a joint mechanism is preferable must include its subsidy and observability. A claim that sequential action improves discovery must report expected rounds and actions. The relevant architecture is a tuple, not a label.

For the Common-Source theorem, the implication is modest. Acquisition thresholds determine how many independent sources arise under one private reward. They do not determine how those sources are disclosed or allocated. The theorem can be used as an acquisition module inside a richer architecture only after the richer model proves that downstream behavior does not feed back into source payoffs.

12 Experimental predictions

DD-011 translates the project's exact mechanisms into a preregistration-ready synthetic design. The selected design has 20 treatment cells spanning acquisition, attribution, disclosure, action timing, and rewards. It registers eight hypotheses, fifteen outcomes, eight explicit synthetic response scenarios, and six sample sizes. A seeded 384,000-draw Monte Carlo run reports conditional power and minimum detectable effects; a separate verifier recreates every random stream (DD-C-0056).

The source-choice predictions are straightforward to state but remain untested. First, costly independent evidence may be under-acquired relative to assignment in parameter cells near the all-common boundary. Second, assignment should mechanically raise the rate of independent-source use, but its effect on discovery depends on downstream action allocation. Third, source provenance may help participants interpret reports, yet it need not increase truthful reporting without attribution or verification. Fourth, acquisition and marginal-coverage allocation may be complementary because the former creates diverse evidence and the latter rewards diverse actions.

The exact model does not supply behavioral effect sizes. DD-011 therefore uses declared synthetic assumptions rather than treating theorem differences as human treatment effects. Under its rational synthetic scenario, all eight assumed effects reach estimated power at least 0.80 by total sample size 960, while 140 of 192 rows at sample sizes 640 or greater remain retained calibration failures across

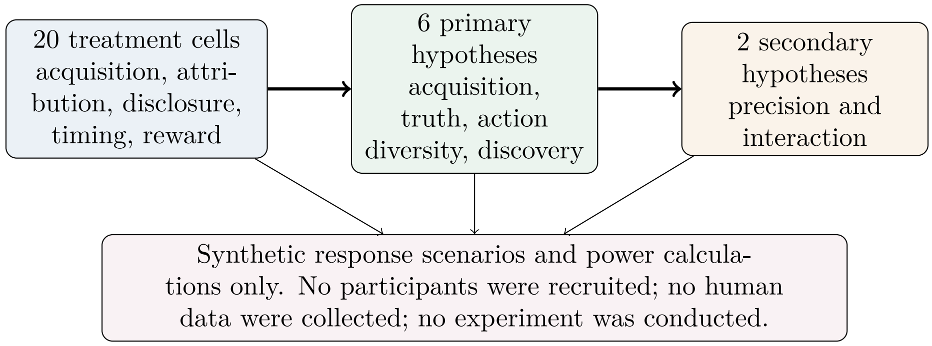

Figure 9: DD-011 maps exact-model mechanisms into experimental predictions without turning them into empirical effects. The design is preregistration-ready, not preregistered or deployed.

Source runs: `20260721T185647Z_DD-011_fa0271d9_fcaa647c55`; claims: DD-C-0056; generator: `distributed_discovery.papers.build_common_source_trap`; input SHA-256 prefixes: `0d586f4001c8`.

all scenarios. Those failures are evidence about sensitivity of the simulation design, not evidence that a real experiment is infeasible.

Several observations would be especially informative. If participants acquire more independence than the planner benchmark in high-accuracy interior cells, that would echo DD-C-0058 rather than falsify the entire framework. If public disclosure raises agreement but not discovery, that would support separating information and allocation. If a joint mechanism improves reports but not obedience, it would locate the failure at the action layer. If provenance changes confidence but not source choice, it would separate interpretability from acquisition incentives.

Boundary. No participants were recruited. No human data were collected. No experiment was conducted. Separate ethics and institutional review are required before deployment. The materials are an unsubmitted preregistration-ready kit, not a preregistration or a claim of empirical power.

13 Limitations

The first limitation is external validity. Every result in the core model is mathematical evidence about a synthetic finite game. No dataset establishes that research teams, firms, investors, or multi-agent systems face homogeneous three-target signals, equal prize splitting, or direct clue-following. The model is diagnostic: it shows that a source-choice wedge can arise under transparent conditions and identifies the precise margin responsible.

Second, sources are stylized. All signals share accuracy p, incorrect labels are symmetric, and independent sources are conditionally independent given the target. Real sources can share training data, incentives, measurement devices, or preprocessing pipelines. Nominally different providers may therefore be one effective channel. Conversely, one provider may deliver conditionally diverse measurements. Extending the theorem to heterogeneous accuracy or arbitrary dependence requires a new model and may destroy the scalar count representation.

Third, action is mechanical. Each agent searches its clue directly. There is no strategic report, no communication, no role policy, and no ability to choose a portfolio conditional on all signals. This makes source effects identifiable inside the model, but it excludes the allocation mechanisms that motivate the broader distributed-discovery program. The Atlas examples show that independent evidence can be wasted by consensus or improved by sequential/differentiated action.

Fourth, the reward is a single equal-split prize. Prize competition creates the interior over-acquisition counterexample. Other sharing rules, individual action values, risk preferences, budgets, and source ownership can change both private thresholds and the planner objective. The general theorem is general in finite N only after all those primitives are frozen.

Fifth, equilibrium is pure and anonymous by source count. Mixed source choice, correlated strategies, asymmetric costs, coalition deviations, and dynamic learning are outside the analysis. Threshold ties are retained, but no selection rule predicts which weak equilibrium occurs. Finite-grid gap tables use an explicit max-planner/min-equilibrium convention and should not be read as frequency predictions.

Sixth, welfare counts union discovery minus acquisition costs. The prize is treated as a transfer. Source acquisition may produce reusable knowledge, spillovers, robustness, calibration, or option value not captured by immediate target discovery. Common sources may support compatibility or coordinated execution. Independent sources may impose integration costs. These additions could reverse planner comparisons.

Seventh, the institutional evidence comes from different registered fixtures. DD-006B, DD-009, and DD-011 do not form one nested model. Their values cannot be added into a single treatment effect or welfare calculation. The figures use them to map margins and feasibility boundaries, not to claim a globally optimal institution.

Finally, the literature review is targeted rather than systematic. It establishes substantial prior art and supports conservative positioning; it does not prove that no mathematically equivalent threshold has appeared elsewhere. The project makes no verified novelty claim. "Common-Source Trap" is a descriptive label for the registered mechanism, not ownership of a field or universal phenomenon.

14 Conclusion

The Common-Source Trap has a precise general boundary. In the frozen finite-team model, a prospective independent-source buyer compares two expected reciprocal prize shares. Those gross gains form a strictly decreasing sequence, yielding a complete weak pure equilibrium-count characterization. Planner gains are the declining increments of union discovery. At the first-source boundary, the two sequences differ by exactly $p(1-p)/N$.

That formula sharpens the two-agent intuition. An agent that purchases the first independent source pays the whole cost but captures only part of the new union probability because correct agents split the prize. The planner values the full new route to discovery. For costs between the thresholds, all-common use can be an equilibrium even though every planner optimum includes independence.

The result is also narrower than the label may suggest. Private thresholds do not remain below planner thresholds at every interior count. The exact $N = 3$, $p = 4/5$ fixture produces more equilibrium independence than the planner wants. Any policy or empirical design built from the theorem must therefore measure the relevant margin rather than assume that "more independent sources" is always the corrective direction.

The institutional implication is layered. Acquisition determines whether distinct evidence exists. Disclosure determines who can use it. Allocation determines whether evidence becomes a diversified portfolio of actions. Rewards and observability determine whether reports and actions follow the intended protocol. The project evidence contains exact bounded witnesses at each layer, but no single mechanism solves all layers under all constraints.

The practical research standard is therefore explicitness: declare the source law, cost, information boundary, action technology, reward, equilibrium selection, and social objective; preserve exact and negative results; and test each intervention at the layer it changes. Under that discipline, the Common-Source Trap is neither a slogan that common information is bad nor a universal under-acquisition theorem. It is an auditable mechanism linking private source choice to collective discovery. In particular, it is not a universal under-acquisition theorem.

A Proof details

A.1 Derivation of the private threshold

Fix $k < N$ and let $m = N - k$. Before switching, the prospective buyer is a common user. Conditional on a correct common draw and $X_k = x$ correct independent users, its prize share is $1/(m+x)$, yielding equation (2). After switching, the agent has an independent draw and there remain $m - 1$ common users. Conditional on the buyer being correct, the common draw is wrong with probability q and correct with probability p. Its gross payoff is therefore

$$I_{k+1} = p\left[q\mathbb{E}\frac{1}{1+X_k} + p\mathbb{E}\frac{1}{m+X_k}\right].$$

Subtracting $C_k = p\mathbb{E}[1/(m + X_k)]$ gives

$$I_{k+1} - C_k = pq\mathbb{E}\left[\frac{1}{1+X_k} - \frac{1}{m+X_k}\right],$$

which is equation (10).

A.2 Positive-kernel identity

For $m \geq 2$, write $X_{k+1} = X_k + Y$ with independent $Y \sim \text{Bernoulli}(p)$. The difference between adjacent normalized threshold kernels, conditional on $X_k = x$, is

$$\begin{aligned}
&f_m(x) - qf_{m-1}(x) - pf_{m-1}(x+1) \\
&= \left(\frac{1}{x+1} - \frac{1}{x+m}\right) - q\left(\frac{1}{x+1} - \frac{1}{x+m-1}\right) - p\left(\frac{1}{x+2} - \frac{1}{x+m}\right).
\end{aligned}$$

Putting the terms over the common denominator $(x+1)(x+2)(x+m-1)(x+m)$ gives the numerator

$$(x+1)(x+2) + p(m-2)(2x+m+1).$$

Every factor is nonnegative and $(x+1)(x+2) > 0$, proving strict positivity. Taking expectations and multiplying by $pq > 0$ establishes $A_k > A_{k+1}$. At $m = 1$, the two reciprocal terms in equation (10) are equal, so $A_{N-1} = 0$.

A.3 Planner concavity and count characterization

For $k < N - 1$, equation (1) gives

$$G_N(k+1, p) - G_N(k, p) = q^{k+1} - q^{k+2} = pq^{k+1}.$$

For $k = N - 1$, both gross values equal $1 - q^N$. Hence the gross increments B_k decline strictly until the terminal zero. Net increments are $B_k - c$. A count maximizes W_N exactly when the preceding net increment is weakly positive and the following net increment is weakly negative, which yields the planner threshold characterization.

A.4 Large-team limit

The binomial reciprocal identity follows by integration:

$$\begin{aligned}\mathbb{E}\frac{1}{1+X_k} &= \sum_{x=0}^{k}\binom{k}{x}p^x q^{k-x}\int_0^1 t^x\,dt \\ &= \int_0^1 (q+pt)^k\,dt = \frac{1-q^{k+1}}{(k+1)p}.\end{aligned}$$

For fixed k, the second expectation in equation (10) is bounded by $1/(N-k)$ and vanishes as N grows. Equation (14) follows.

B Reading the finite tables

The generated grids fix $p = 2/3$, use the registered costs $0, 1/24, 1/12, 1/8, 1/6, 1/4$, and display $N = 2, \ldots, 8$. They are slices of the full 126-row DD-008A census. Each row contains every count $k = 0, \ldots, N$ with gross discovery, net value, weak/strict equilibrium flags, and planner optima.

The equilibrium figure reports the minimum weak-equilibrium count. The planner figure reports the maximum planner count. These conventions reproduce the registered independence-gap statistic. At $c = 0$, both $N-1$ and N can be optimal or weak equilibria because the last transition changes neither signals nor discovery and costs nothing. The max/min convention can therefore display a count gap even when the underlying gross and net values tie.

The discovery-gap figure evaluates gross discovery at those displayed counts. The welfare-gap figure evaluates net value. Because the planner count is chosen from the exact argmax set, the welfare difference is never negative. The discovery difference can be negative when a privately chosen extra source raises coverage by less than its cost. Retaining those cells is essential to the paper's negative result.

No color scale, interpolation, smoothing, or estimated surface is applied to the finite grids. Values are exact integers or fractions copied only after manifest checksum validation. The trap-region curves are evaluations of the analytic formulas, not fitted curves through the finite census.

C Reproducibility and evidence provenance

The source repository builds this paper with `distributed_discovery.papers.build_common_source_trap`. Before generating an asset, the builder loads each immutable run manifest, requires passed validation and exit status zero, resolves the declared output, and recomputes its SHA-256 checksum. A mismatch aborts the build.

Generated TeX assets contain source comments, run IDs, claim IDs, the generator name, and input checksum prefixes. The machine-readable provenance file records complete hashes for all source artifacts and generated assets. Citation keys are resolved against the repository bibliography, and every DD-C identifier is resolved against the claim ledger. The builder rejects missing required sections, unresolved references or citations, and overfull boxes.

Tectonic compiles the document twice in isolated temporary directories with a fixed `SOURCE_DATE_EPOCH` derived from the first source run. The two PDF byte strings must have identical SHA-256 checksums. The validated PDF, build log, page count, checksum, and source-run mapping are committed. A separate Poppler workflow renders every page to PNG for visual inspection; any PDF checksum change invalidates that record.

The public site copies the PDF only when its checksum matches the validation record. The publication page reports the checksum and working-paper status. No DOI, archival deposit, journal submission, or peer-review status is asserted. Generated site output remains uncommitted and is rebuilt from main by GitHub Actions.

D Extension audit

The scalar threshold theorem is deliberately modular. An extension should state which primitive changes and then re-establish only the identities that depend on that primitive. This appendix records the minimum audit needed before carrying the paper's language into a richer model.

Changed primitive	Result that must be re-derived	Diagnostic that remains useful
Heterogeneous source accuracy	Individual common-to-independent gains; a count alone may no longer classify profiles.	Enumerate unilateral deviations and check prize conservation state by state.
Correlated independent sources	Union discovery G_N and planner increments B_k; conditional independence no longer factors failures.	Compare the private gain from a new route with its exact marginal union coverage.
Alternative prize sharing	Private thresholds A_k, including monotonicity; the planner sequence is unchanged only if rewards remain transfers.	Separate gross discovery, private rewards, and real resource costs.
Strategic reports or actions	Source payoffs depend on continuation behavior; truth and obedience become equilibrium constraints.	Audit acquisition, reporting, and action deviations as distinct layers.
Dynamic acquisition	One-shot source counts are replaced by histories, beliefs, and continuation values.	Preserve the public-good comparison while testing free-riding and encouragement effects separately.
Reusable evidence or robustness value	The planner objective must include spillovers beyond immediate union discovery.	Report each social objective and its units rather than combining incomparable benefits.

Table 2: Minimum proof obligations for common extensions. The right column is a diagnostic workflow, not a claim that the present theorem survives the change.

Three promotion rules follow. First, a replicated numerical pattern is not an analytic extension until the new payoff difference is derived and its sign is proved on a declared domain. Second, a richer mechanism result does not repair source acquisition unless source purchase is endogenous in that mechanism. Third, an empirical treatment effect cannot validate the theorem without a measurement model connecting observed choices to its source, signal, and reward primitives. These rules preserve the distinction among exact evidence, synthetic design evidence, and future human evidence.

The most direct theoretical extension is heterogeneous accuracy with observable provider labels. It would replace the anonymous count k by a profile of source qualities and could reveal sorting or asymmetric free-riding. The most direct institutional extension combines verifiable source assignment with the joint truth-and-allocation mechanism, including a balanced-budget constraint. The most direct empirical extension uses DD-011's preregistration-ready cells after ethics review, pilot calibration, and an explicit mapping from experimental actions to source independence. None of those extensions is executed or claimed here; the table makes their proof and evidence obligations inspectable.

References

Jason McKenzie Alexander, Johannes Himmelreich, and Christopher Thompson. Epistemic landscapes, optimal search, and the division of cognitive labour. *Philosophy of Science*, 82(3):424–453, 2015. doi: 10.1086/681766.

S. Nageeb Ali. Herding with costly information. *Journal of Economic Theory*, 175:713–729, 2018. doi: 10.1016/j.jet.2018.02.009.

Patrick Bolton and Christopher Harris. Strategic experimentation. *Econometrica*, 67(2):349–374, 1999. doi: 10.1111/1468-0262.00022.

Luca Colombo, Gianluca Femminis, and Alessandro Pavan. Information acquisition and welfare. *The Review of Economic Studies*, 81(4):1438–1483, 2014. doi: 10.1093/restud/rdu015.

Christian Hellwig and Laura Veldkamp. Knowing what others know: Coordination motives in information acquisition. *The Review of Economic Studies*, 76(1):223–251, 2009. doi: 10.1111/j.1467-937X.2008.00515.x.

Jack Hirshleifer. The private and social value of information and the reward to inventive activity. *American Economic Review*, 61(4):561–574, 1971. URL `https://www.jstor.org/stable/1811850`.

Lu Hong and Scott E. Page. Groups of diverse problem solvers can outperform groups of high-ability problem solvers. *Proceedings of the National Academy of Sciences*, 101(46):16385–16389, 2004. doi: 10.1073/pnas.0403723101.

Godfrey Keller, Sven Rady, and Martin Cripps. Strategic experimentation with exponential bandits. *Econometrica*, 73(1):39–68, 2005. doi: 10.1111/j.1468-0262.2005.00564.x.

Philip Kitcher. The division of cognitive labor. *The Journal of Philosophy*, 87(1):5–22, 1990. doi: 10.2307/2026796.

Annie Liang and Xiaosheng Mu. Complementary information and learning traps. *The Quarterly Journal of Economics*, 135(1):389–448, 2020. doi: 10.1093/qje/qjz033.

David P. Myatt and Chris Wallace. Endogenous information acquisition in coordination games. *The Review of Economic Studies*, 79(1):340–374, 2012. doi: 10.1093/restud/rdr018.

Andrey Rzhetsky, Jacob G. Foster, Ian T. Foster, and James A. Evans. Choosing experiments to accelerate collective discovery. *Proceedings of the National Academy of Sciences*, 112(47):14569–14574, 2015. doi: 10.1073/pnas.1509757112.

Leonid Tiokhin and Maxime Derex. Competition for novelty reduces information sampling in a research game: A registered report. *Royal Society Open Science*, 6(5):180934, 2019. doi: 10.1098/rsos.180934.

Michael Weisberg and Ryan Muldoon. Epistemic landscapes and the division of cognitive labor. *Philosophy of Science*, 76(2):225–252, 2009. doi: 10.1086/644786.

Kevin J. S. Zollman. The epistemic benefit of transient diversity. *Erkenntnis*, 72(1):17–35, 2010. doi: 10.1007/s10670-009-9194-6.